Arista Access Node KR-ACC-82 Optical Transport Failure SOP

Node: KR-ACC-82 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_8286

Description

This Standard Operating Procedure (SOP) is designed to guide the resolution of a Critical Optical Transport Failure on the Arista Access Node KR-ACC-82, located at KR Puram Access. The failure domain encompasses SFP/XFP degradation, fiber cuts, and Rx/Tx power loss. This failure occurs due to the inherent vulnerability of optical interfaces to environmental stressors and physical damage. The Arista Access Node, operating in a role-specific capacity, relies heavily on the integrity of its optical transport mechanisms to ensure seamless data transmission. The SFP/XFP modules, responsible for converting electrical signals to optical signals, are prone to degradation over time due to factors such as high temperatures, excessive humidity, and physical stress. Additionally, fiber cuts and Rx/Tx power loss can compromise the optical transport chain, hindering data transmission. Therefore, it is imperative to promptly diagnose and resolve such failures to prevent network outages and ensure uninterrupted service.

Pre-requisite Checks

- check: Verify that remote hands have been dispatched to KR Puram Access
- check: Confirm that redundant link status is operational
- check: Ensure that all necessary safety protocols have been implemented

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show interfaces ae0/0, Expected Output: Arista EOS, Description: This command displays the interface status, including optical signal quality and SFP/XFP temperature. It helps isolate potential issues related to SFP/XFP degradation or fiber cuts.

Command: show hardware sfp ae0/0, Expected Output: SFP information, Description: This command provides detailed information about the SFP/XFP module, including its temperature, voltage, and optical signal quality.

Command: show bgp summary, Expected Output: BGP summary, Description: This command displays BGP peering information, helping to identify
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potential issues related to Rx/Tx power loss or fiber cuts.
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Resolution Steps

Step 1: Step: Replace SFP/XFP module, Command: replace sfp ae0/0 0/0, Description: This command replaces the SFP/XFP module with a new one, resolving issues related to SFP/XFP degradation.

Step 2: Step: Repair or replace fiber, Command: repair fiber ae0/0, Description: This command initiates the repair or replacement of the fiber, resolving issues related to fiber cuts.

Step 3: Step: Adjust Rx/Tx power settings, Command: set interface ae0/0 rx-power -2, Description: This command adjusts the Rx/Tx power settings, resolving issues related to Rx/Tx power loss.

Post-Check Verification

```
Command: show interfaces ae0/0, Expected Output: Arista EOS,
Description: This command verifies that the interface status has been
restored to normal.

Command: show hardware sfp ae0/0, Expected Output: SFP information,
Description: This command verifies that the SFP/XFP module is
functioning correctly.

Command: show bgp summary, Expected Output: BGP summary, Description:
This command verifies that BGP peering information is intact.
```

Roll-back Procedures

- Step: Restore original SFP/XFP module, Command: replace sfp ae0/0 0/0, Description: This command restores the original SFP/XFP module, reverting to the previous state.

- Step: Repair or replace original fiber, Command: repair fiber ae0/0, Description: This command repairs or replaces the original fiber, reverting to the previous state.

- Step: Revert Rx/Tx power settings, Command: set interface ae0/0 rx-power 0, Description: This command reverts the Rx/Tx power settings to their original values.